

Ions and Channels

Computational Neuroscience

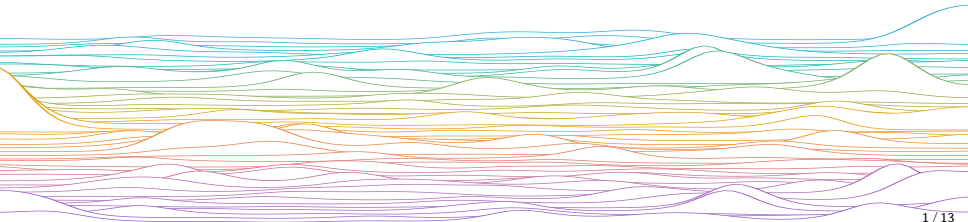
University of Bristol

Adapted from C O'Donnell's slides

M Rule

Learning outcomes:

- ▶ Understand what ion channels are and what they do
- ▶ Write down a mathematical model of an ion channel current
- ▶ Explain how ion channels make the neuron's input-output function nonlinear



Modelling Scales

... Quantum Chemistry, Molecular dynamics

Molecules

Gillespie, Master Equation

Concentrations

Mass-Action Kinetics

Conductance Models

Hodgkin–Huxley

Spiking Models

Leaky Integrate and Fire

Rate Neurons

Neural Mass/Field Models

Poisson Neurons

Generalized Linear Models

Binary Neurons

McCulloch–Pitts, Hopfield, Perceptron

Cognitive Neuroscience, Psychology ...

Physiological,
Quantitative

Biological
Realism,
Data needed
to identify
parameters



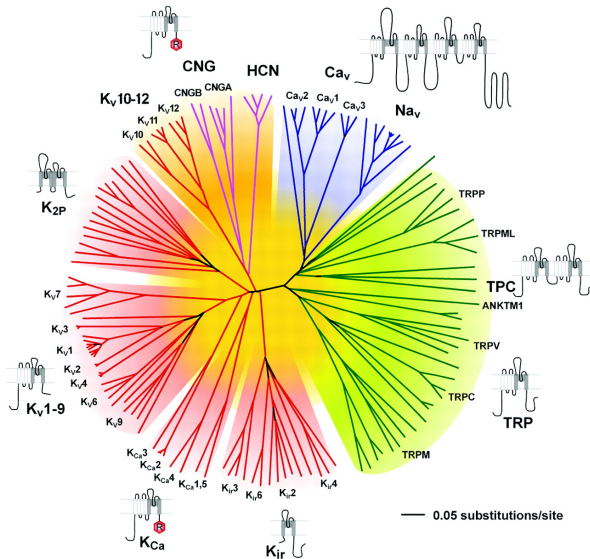
Computational
Efficiency,
Mathematical
Tractability



Phenomenological,
Qualitative

What are ion channels?

- ▶ Ion channels are ion-permeable pores in the lipid membrane of cells.
- ▶ A single neuron typically has hundreds of thousands to millions of ion channels embedded in its membrane. Hille (1992)
- ▶ They open and close in response to stimuli (voltage, neurotransmitters, intracellular chemicals, pH, mechanical forces, temperature...), passing ions like Na^+ , K^+ , Ca^{2+} , Cl^- .
- ▶ Their currents mediate electrical signalling in the nervous system.
- ▶ The conductance of single ion channels vary between 0.1 and 100 picoSiemens. For most channels it's around 10 pS.
- ▶ The flux through a single open channel can be millions of ions per second.



Yu et al., *Pharm Rev* (2005)

Ion channel types

in this course. in biology, much more diversity (eg $\text{Na}_v1.6$)

Sodium (Na^+) channels mediate inward currents that depolarise the voltage.

- ▶ Fast gating, activated by depolarisation (positive feedback).
- ▶ Responsible for upswing of the action potential, and boosting subthreshold inputs in dendrites.
- ▶ Targets for some anaesthetics (e.g. lidocaine, pufferfish venom)

Potassium (K^+)

- ▶ Outward flux of $+$ ions: lowers membrane volates
- ▶ Fast or slow gating, activated by depolarisation (negative feedback).
- ▶ Voltage-independent K^+ channels mediate the 'leak' current.
- ▶ Very genetically diverse (around 50 types in mammals).

Calcium (Ca^{2+}) inward current of positive ions depolarise the voltage.

- ▶ Fast gating, but not as strongly expressed as sodium so have weaker effect on the voltage.
- ▶ Responsible for some forms of dendritic spikes.
- ▶ Generate intracellular calcium signals that the cell uses to monitor its electrical activity.

Other channels include

- ▶ Chloride (Cl^-) channels: involved in setting resting voltage.
- ▶ HCN channels: mixed Na^+/K^+ I_h current

Modelling ion channels

- ▶ In most neuroscience applications we don't care about all the molecular details of the ion channel, we just want a simple model that captures their dynamics.
- ▶ Usually this involves state-based modelling.
- ▶ We assume that each channel can be in one of a small number of discrete states. The channel can transition between states, with transition rates that depend on the cell's voltage.

Modelling ion channels

Consider a 2-state ion channel model with transitions between the closed s_0 and open s_1 states with transition rates α and β .



If we imagine a large population of such channels, we could think of s_1 as representing the proportion of the population in the open state. Then we can write down a differential equation to describe its dynamics:

$$\dot{s}_1(t) = \alpha s_0(t) - \beta s_1(t)$$

The steady state value s_∞ is found by setting $\dot{s}_1 = 0$, so

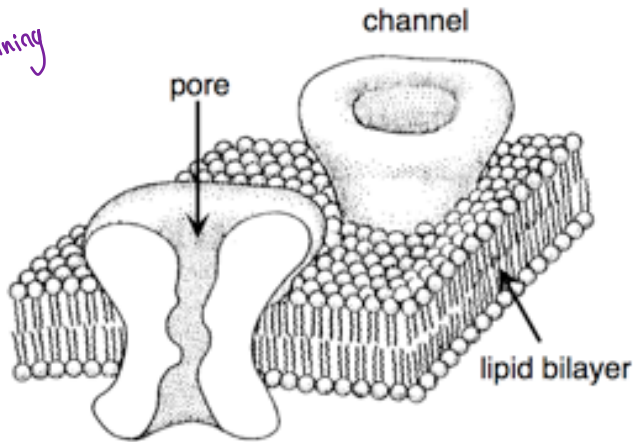
$$s_\infty = \frac{\alpha}{\alpha + \beta}$$

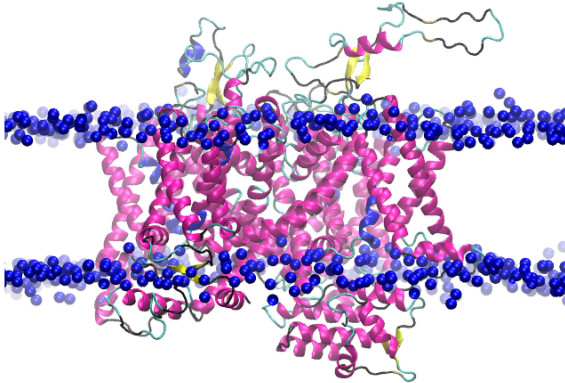
Then we can rewrite the right hand side of the dynamics equations as

$$\dot{s}(t) = \frac{s_\infty - s_1(t)}{\tau}$$

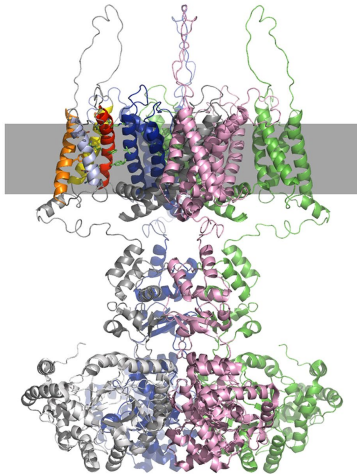
Where we have introduced the time constant $\tau = 1/(\alpha + \beta)$

Todo:
move
to
beginning





Nav1.1; image from Montanino, Annaclaudia, et al. (2020)



Open Kv1.1; image from Hasan, Sonia, et al. (2017)

Modelling ion channels

The electrical current flowing through a large population of such channels is

$$I(t) = \bar{g} \cdot s(t) \cdot [E_{\text{rev}} - v(t)]$$

I am using the electrical engineering and physics current convention here. In electrophysiology the RHS will actually be the negative current. This is a historical artefact related to Hodgkin and Huxley's experimental setup and how they chose current direction conventions in nodal analysis. Hodgkin and Huxley used a voltage-clamp experimental setup where an amplifier in negative feedback supplied current to cancel the synaptic current, holding the cell at a fixed voltage. They measured the output of the amplifier, hence the change in sign convention.

Modelling ion channels

This was a very simple 2-state channel example.

- ▶ Most real channels are too complicated to describe so compactly,
- ▶ so their models often have many more states.

The voltage dependence is built into these channel models by making the the transitions rate (α , β) functions of voltage.

We will go through a famous example of the this in the next video: the Hodgkin-Huxley squid axon model.

You can find lots of example computational models of ion channel types in several good online repositories:

[ModelDB](#)

[Channelpedia](#)

[ICGenealogy](#)

end